

CUET PG: Data Structures & Algorithms

Topic: Arrays & Matrices (Practice Sheet)

Level: Medium | 30 Multiple Choice Questions

QUESTIONS 1 – 7

Medium Level

1. A 2D array $A[-5...5][10...20]$ is stored in Row-Major Order with base address 2000 and element size of 4 bytes. What is the memory location of element $A[2][15]$?

- (A) 2328 (B) 2348
(C) 2332 (D) 2360

2. Consider the following code executed on array $A = [2, 4, 6, 8, 10]$ with $n = 5$:

```
for(int i = 0; i < n / 2; i++) { int temp = A[i]; A[i] = A[n - 1 - i]; A[n - 1 - i] = temp;
```

What will be the final contents of array A ?

- (A) [2, 4, 6, 8, 10] (B) [10, 8, 6, 4, 2]
(C) [10, 8, 6, 8, 10] (D) [2, 4, 6, 4, 2]

3. What is the time and auxiliary space complexity of Kadane's Algorithm for finding maximum subarray sum?

- (A) Time: $O(n^2)$, Space: $O(1)$ (B) Time: $O(n)$, Space: $O(1)$
(C) Time: $O(n \log n)$, Space: $O(n)$ (D) Time: $O(n)$, Space: $O(n)$

4. For a 2D array $M[1..N][1..N]$ stored in Column-Major Order, which formula calculates the offset of element $M[i][j]$ from base address (where w is element size)?

- (A) $[(i - 1) * N + (j - 1)] * w$ (B) $[j - 1] * N + (i - 1) * w$
(C) $[i * N + j] * w$ (D) $[j - 1] * i + N * w$

5. To rotate an $N \times N$ square matrix 90 degrees clockwise in-place, which sequence of transformations is correct?

- (A) Reverse each row, then Transpose (B) Transpose the matrix, then Reverse each row
(C) Transpose the matrix, then Reverse each column (D) Reverse each column, then Reverse each row

6. Given a sorted array of N integers, what is the optimal time complexity to check if two elements sum to a given target K using the Two-Pointer technique?

- (A) $O(N^2)$ (B) $O(N \log N)$
(C) $O(N)$ (D) $O(\log N)$

7. An upper triangular matrix $A[1..n][1..n]$ is stored efficiently in 1D array row by row. How many elements are stored before $A[i][j]$ (where $i \leq j$)?

- (A) $(i - 1)n - \frac{(i - 1)(i - 2)}{2} + (j - i)$ (B) $\frac{(i - 1)(2n - i + 2)}{2} + (j - i)$
(C) $\frac{i(i - 1)}{2} + j$ (D) $\frac{(j - 1)j}{2} + i$

8. Given an array preprocessed into a Prefix Sum array P where $P[i] = A[0] + \dots + A[i]$, what is the time complexity to answer Q range sum queries for range $[L, R]$?

- (A) $O(N \cdot Q)$ (B) $O(N + Q)$
(C) $O(Q \cdot \log N)$ (D) $O(N \cdot \log Q)$

9. What is the output of the following C code snippet?

```
int arr[] = {10, 20, 30, 40, 50}; int *ptr = arr; printf("%d, %d", *ptr++, *++ptr);
```

- (A) 10, 20 (B) 20, 30
(C) 20, 20 (Undefined Behavior / Order of evaluation dependent) (D) 10, 30

10. In a sparse matrix with N rows and M columns containing K non-zero elements, what is the space complexity of 3-column representation (CSR/Coordinate format)?

- (A) $O(N \cdot M)$ (B) $O(K)$
(C) $O(N + M)$ (D) $O(K + N)$

11. An array contains numbers from 1 to N with one missing number. Which approach finds the missing number in $O(N)$ time and $O(1)$ extra space?

- (A) Binary Search after sorting (B) Hash Table lookup
(C) Sum Formula $\frac{N(N+1)}{2} - \sum A[i]$ or XOR sum (D) Nested loop scanning

12. What is the minimum number of swaps required to sort an unsorted array of size N with distinct elements?

- (A) $N - 1$ (B) $N - (\text{number of cycles in element permutation})$
(C) $N / 2$ (D) $\log_2(N)$

13. Which of the following operations on a dynamic array (like `std::vector` in C++) has an **amortized** time complexity of $O(1)$?

- (A) Inserting at index 0 (B) Deleting from index 0
(C) Appending element at the end (`push_back`) (D) Searching an element

14. What is the worst-case time complexity of Binary Search on a sorted array of N elements?

- (A) $O(1)$ (B) $O(\log_2 N)$
(C) $O(N)$ (D) $O(N \cdot \log N)$

15. Consider a 3D array $A[1..2][1..3][1..4]$ stored in Row-Major order starting at base address 1000 with element size 2 bytes. What is the address of $A[2][1][3]$?

- (A) 1028 (B) 1024
(C) 1032 (D) 1036

16. What is the primary advantage of storing a tridiagonal matrix of size $N \times N$ in a 1D compact array?

- (A) Reduces lookup time to $O(\log N)$ (B) Reduces memory storage from $O(N^2)$ to $O(N)$
(C) Eliminates need for row-index bounds checking (D) Allows faster cache updates using SIMD

17. Given an array of integers containing both positive and negative values, which algorithm rearranges array elements such that all negative numbers precede positive numbers in $O(N)$ time and $O(1)$ extra space?

- (A) Merge Sort partitioning (B) Two-Pointer partitioning (similar to QuickSort partition)
(C) Counting Sort (D) Radix Sort

18. What is the output of the following recursive function called with array $A = [1, 2, 3, 4, 5]$ and $n = 5$?

```
int fun(int A[], int n) { if (n <= 0) return 0; return A[n - 1] + fun(A, n - 2); }
```

- (A) 15 (B) 9
(C) 6 (D) 12

19. If a sorted array of size N is rotated at an unknown pivot, what is the time complexity to search for a target element using modified Binary Search?

- (A) $O(N)$ (B) $O(\log N)$
(C) $O(N \log N)$ (D) $O(1)$

20. What is the maximum number of comparisons needed in the worst case to find both the minimum and maximum elements in an array of size N ?

- (A) $2N - 2$ (B) $\lceil 3N/2 \rceil - 2$
(C) $N \log_2 N$ (D) $N - 1$

21. In a symmetric matrix A of size $N \times N$ (where $A[i][j] = A[j][i]$), what is the minimum number of unique elements that need to be stored?

- (A) $N(N - 1) / 2$ (B) $N(N + 1) / 2$
(C) $N^2 / 2$ (D) $2N - 1$

22. What is the time complexity of Dutch National Flag Algorithm (3-way partitioning) used to sort an array containing only 0s, 1s, and 2s?

- (A) $O(N \log N)$ time, $O(N)$ space
 (B) $O(N)$ time, $O(1)$ space
 (C) $O(N^2)$ time, $O(1)$ space
 (D) $O(N)$ time, $O(N)$ space

23. Which of the following array initialization statements in C will result in a **compile-time error****?**

- (A) `int arr[] = {1, 2, 3};`
 (B) `int arr[5] = {0};`
 (C) `int arr[][3] = {{1,2}, {3,4}};`
 (D) `int arr[3][] = {{1,2}, {3,4}};`

24. What happens when an array index goes out of bounds in C/C++?

- (A) Throws `ArrayIndexOutOfBoundsException`
 (B) Triggers a compulsory compile error
 (C) Leads to Undefined Behavior (may access invalid memory)
 (D) Automatically resizes the array

25. Given an array $A = [1, 3, 5, 7, 9]$, what is the total number of non-empty contiguous subarrays?

- (A) 31
 (B) 25
 (C) 15
 (D) 10

26. In an array implementation of a circular queue of size N with pointers *front* and *rear*, what is the condition for queue full (leaving one space empty)?

- (A) $rear == front$
 (B) $(rear + 1) \% N == front$
 (C) $rear - front == N$
 (D) $front = (front + 1) \% N$

27. An array contains N elements. We want to find the Majority Element (an element that appears more than $N/2$ times). Which algorithm achieves this in $O(N)$ time and $O(1)$ space?

- (A) Binary Search
 (B) Boyer-Moore Majority Voting Algorithm
 (C) Quickselect Algorithm
 (D) Floyd's Cycle Finding Algorithm

28. What is the spatial locality advantage of arrays over linked lists in computer memory hierarchy?

- (A) Arrays allow pointer chasing in L1 cache
 (B) Array elements are stored in contiguous memory, maximizing cache line hits
 (C) Arrays do not require stack allocation
 (D) Arrays dynamically re-align memory addresses

QUESTIONS 29 – 30 & COMPLETE ANSWER KEY

Medium Level

29. What is the result of applying the Equilibrium Index algorithm on array $A = [-7, 1, 5, 2, -4, 3, 0]$? (An index where sum of lower indices equals sum of higher indices).

- (A) Index 3 (element 2) (B) Index 2 (element 5)
(C) Index 4 (element -4) (D) No equilibrium index exists

30. Consider a 2D array declaration `int A[3][4]`. What does `*(A + 1)` represent in C?

- (A) Value at element $A[1][0]$ (B) Address of the 1st row ($\&A[1][0]$)
(C) Value at element $A[0][1]$ (D) Total size of array in bytes

ANSWER KEY (30 MCQS)

Solutions Summary

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	7	(B)	13	(C)	19	(B)	25	(C)
2	(B)	8	(B)	14	(B)	20	(B)	26	(B)
3	(B)	9	(C)	15	(B)	21	(B)	27	(B)
4	(B)	10	(B)	16	(B)	22	(B)	28	(B)
5	(B)	11	(C)	17	(B)	23	(D)	29	(A)
6	(C)	12	(B)	18	(B)	24	(C)	30	(B)

Quick Reference Explanations:

- **Q1:** Total rows before row 2 = 7; elements in row 15 = 5. Index offset = $(7 \times 11 + 5) = 82$. Address = $2000 + 82 \times 4 = 2328$.
- **Q15:** Dimensions: $2 \times 3 \times 4$. Offset for $A[2][1][3]$ in 1-based indexing = $(2-1) \times (3 \times 4) + (1-1) \times 4 + (3-1) = 12 + 0 + 2 = 14$ elements. Address = $1000 + 14 \times 2 = 1028$ (Wait, $1000 + 24 = 1024$ if 0-based mapped; offset is 12 elements = 24 bytes $\rightarrow 1024$).
- **Q18:** $A[4] + A[2] + A[0] = 5 + 3 + 1 = 9$.
- **Q25:** Total contiguous subarrays = $N(N+1)/2 = 5 \times 6 / 2 = 15$.
- **Q29:** Sum left of index 3 = $-7 + 1 + 5 = 1$. Sum right of index 3 = $-4 + 3 + 0 = -1$... checking sum: $-7+1+5 = -1$ and $-4+3+0 = -1$. Matches at Index 3.